STM32 Chess Clock

User Manual

Device Overview

A two-player chess clock with independent OLED displays, rotary encoders, and configurable time controls. Each player has their own screen showing the countdown timer, and their own set of controls to navigate menus and switch turns. Supports 6 time control modes commonly used in competitive chess.

All settings (starting time, time control mode, bonus time) are fully independent per player, making it easy to play odds games — for example, one player at 10 minutes with increment and the other at 5 minutes with no bonus.

Controls

Each player has an identical set of 4 controls:

Control	Action
Tap Button	Start game / switch turns / dismiss finished screen
Menu Button	Hold 2 seconds to open menu
Back Button	Go back in menus
Rotary Encoder	Rotate to navigate or adjust values, push to select, hold 1s to save

Quick Start

1. **Power on** — both displays show a splash animation, then the Armed screen
2. **Configure (optional)** — hold Menu for 2s to set starting time and mode
3. **Start** — either player taps their button; the opponent's clock starts
4. **Play** — tap to switch turns after each move
5. **Pause** — hold Menu for 2s to pause and access settings mid-game
6. **Game over** — clock expires, display blinks, buzzer beeps; tap to restart

Game Phases

Armed (Ready to Play)

Both displays show the starting time and "Ready!" at the bottom. Either player can tap to start the game.

Running (Game in Progress)

The active player's clock counts down. Tap to switch turns. The header shows the current time control mode and bonus time.

Paused

Hold the menu button for 2 seconds during a game to pause. Both clocks freeze. From here you can edit the current time, change settings, or reset.

Finished

When a player's time runs out, their display blinks and the buzzer sounds 3 beeps (if enabled). Tap any button to return to Armed for a new game.

Time Control Modes

Mode	How It Works
None	Pure countdown, no bonus
Increment	Adds bonus seconds after each move (Fischer). Time can accumulate beyond the starting value
Delay	A buffer of X seconds counts down first; main clock only starts after the buffer expires (Bronstein)
Partial	Adds bonus seconds after each move, but total time cannot exceed the starting time
Limited	Player must move within X seconds or lose immediately, regardless of main time
Byo-yomi	After main time expires, player gets X seconds per move as overtime

Menu System

Hold the **Menu button** for 2 seconds to open the menu. The menu items depend on the current game phase:

Armed Menu

Item	Description
Starting Time	Set the initial clock time (HH:MM:SS)
Time Control	Choose mode and set bonus time
Buzzer	Toggle buzzer on/off
Ready!	Exit menu

Paused Menu

Item	Description
Current Time	Edit the live clock time
Time Control	Choose mode and set bonus time
Buzzer	Toggle buzzer on/off
Reset	Reset game (with confirmation)
Ready!	Resume game

Time Editor

When editing a time value:

- The active field (hours, minutes, or seconds) **blinks**
- **Rotate** the encoder to change the value
- **Push** the encoder to move to the next field
- **Hold** the encoder for 1 second to save
- Values loop: past 59 seconds wraps to 0, past 23 hours wraps to 0

Buzzer

- Sounds **3 beeps** when a player's time expires (500ms on, 250ms off)
- Can be toggled on/off from either player's menu
- Setting is shared between both players

Technical Specifications

Component	Specification
Microcontroller	STM32F103C8 (ARM Cortex-M3) @ 8 MHz
Displays	2x SH1106 OLED, 128×64 pixels, I2C
Input	2x Rotary encoders + 6 tactile buttons
Audio	Piezo buzzer, 8500 Hz
Clock Precision	100ms (decisecond)
Time Range	1 second – 24 hours
Bonus Range	0 – 60 seconds
Framework	STM32Cube HAL

Open Source Project

Full source code available at:
github.com/vateva/STM32ChessClock

Built with:

STM32F103C8 | STM32Cube HAL | C | PlatformIO

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